Machine Learning

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eg. Email spam classification

email text → [model] → spam (-ve) / ham (+ve)

spam → email with code, virus

ham → imp / safe email

safe email
Ham email
↓
ML model
- spam → spam folder → missed an imp email → moderately safe
- ham → inbox → completely safe

spam email
↓
ML model
- spam → spam folder → completely safe
- ham → inbox → virus on system or lot of code in inbox → very unsafe

In above case, if model is predicting any email as ham then we want most of them to be correct predictions

$y_{pred} = +ve$
↓
TP + FP

$y_{pred} = y_{act}$
↓
TP

In short we are trying to maximize TP within TP + FP that is maximizing precision

$$prec = \frac{TP}{TP + FP}$$

In email clf case precision is preferred over other performance metrics

In multiple cases prec and recall both need to high, in such cases f1 score is used as performance metrics

prec and recall both are equally important

$$f1 = \frac{2}{\frac{1}{prec} + \frac{1}{rec}}$$ harmonic mean

$$\boxed{f1 = \frac{2\, prec \times rec}{prec + rec}}$$

$$f1 = \frac{2pr}{p + r}$$

performance metrics for multiclass classification

height
weight $\longrightarrow$ $\boxed{\text{ML model}}$ $\longrightarrow$ tiger / elephant / horse
T E H

$y_{act} = [T\ T\ E\ E\ H\ H\ H\ T\ E\ H\ E\ T\ H\ T\ E]$

$y_{pred} = [T\ E\ E\ H\ H\ H\ T\ H\ E\ E\ T\ T\ H\ T\ T]$

$$\text{accuracy} = \frac{8}{15}$$

$$\text{prec}(T) = \frac{n[y_{pred} = T \text{ and } y_{pred} = y_{act}]}{n[y_{pred} = T]} = \frac{3}{6}$$

$$\text{prec}(E) = \frac{n[y_{pred} = E \text{ and } y_{pred} = y_{act}]}{n[y_{pred} = E]} = \frac{2}{4}$$

$$\text{prec}(H) = \frac{n[y_{pred} = H \text{ and } y_{pred} = y_{act}]}{n[y_{pred} = H]} = \frac{3}{5}$$

A good multiclass clf model should have high precision for every category

$$\text{rec}(T) = \frac{n[y_{pred} = T \text{ and } y_{pred} = y_{act}]}{n[y_{act} = T]} = \frac{3}{5}$$

$$\text{rec}(E) = \frac{2}{5} \qquad \text{rec}(H) = \frac{3}{5}$$

$$\text{f1 score}(T) = \frac{2pr}{p+r} = \frac{2 \times 3/6 \times 3/5}{\frac{3}{6} + \frac{3}{5}}$$

$0 \leq \text{prec} \leq 1 \qquad 0 \leq \text{rec} \leq 1 \qquad 0 \leq \text{f1 score} \leq 1$

to compute overall prec / rec / f1 score of model we take weighted average

$$prec = \frac{n[Y_{act} = T] \times prec(T) + n[Y_{act} = E] \times prec(E) + n[Y_{act} = H] \times prec(H)}{n[Y_{act}]}$$

Classes = $C = [T, E, H]$ $\quad m = 3$

$$prec = \frac{\sum_{i=1}^{m} n[Y_{act} = C_i] \times prec(C_i)}{n[Y_{act}]}$$

$$rec = \frac{\sum_{i=1}^{m} n[Y_{act} = C_i] \times rec(C_i)}{n[Y_{act}]}$$

for last example

$$\text{model recall} = \frac{\frac{3}{5} \times 5 + \frac{2}{5} \times 5 + 3/5 \times 5}{15}$$

$$\text{model prec} = \frac{3/6 \times 5 + 2/4 \times 5 + 3/5 \times 5}{15}$$

model f1 = use f1 formula